CitiServe: A Mobile Platform for Efficient Information and Service Dissemination in Local Government Units

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Chapter 1

INTRODUCTION

1.1 Background of the Study

As the modern world continues to advance, information is rapidly increasing in value. Information is any knowledge acquired through research or analysis that is passed from one person to another through different media [1]. It can exist in numerous forms, such as numerical data, messages, signals, intelligence, or facts. Individuals, organizations, and societies depend heavily on information, and there is a constant need for it in almost all fields [2]. In order to make decisions that will improve lives and enterprises, having access to timely and accurate information in a fast and efficient way is a must [2]. Advancements in medicine, economy, technology, education, and many more are driven by information.

As of July 2024, the total users of the internet around the world have reached 5.45 billion, which is equivalent to 67.1 percent of the world's population [3]. This means that information, better known as data, is very prevalent and reaches almost everyone in the world. The widespread circulation of information is possible through information dissemination, which allows information to reach multiple audiences. A literature review by Liah Shonhe [4] showed that advancements in ICTs are key points in improving access, efficiency, and innovativeness. Technology can utilize different techniques such as QR codes, social networks, social media, internet websites, and digital repositories.

The Philippines implements several of these techniques especially social media, being regarded as the social media capital of the world [5]. With a total population of 118.2 million, 86.75 million, or 73.4 percent are active social media users. An average of 3 hours

and 34 minutes is spent by Filipinos using social media. A study that focuses on the emergent modes of information dissemination in a senior high school department [6], found that SMS notifications were the best mode to seek and provide information when urgency was a big factor. This study utilized five categories to determine the degree of significance of the emerging modes and showed that in almost all categories, SMS notifications were the most common mode. A study evaluated the information-dissemination strategy used by the Barangay Masili health clinic [7]. Their findings demonstrated that door-to-door, or face-to-face, contact by BHWs was the most often used modality by the barangay. This study highlights how effective and efficient information and service dissemination is important even at the barangay level.

With this, the researchers realized that there is a need to improve how information and services are disseminated to citizens in an effective, timely, and efficient way. A way to address this is by creating a mobile platform that every citizen can have access to.

1.2 Statement of the Problem

This study aims to develop a mobile platform that serves as a centralized information hub for accessing government information and services, thus reducing reliance on manual methods; and evaluate the effectiveness of this mobile platform in enhancing communication between the local government and its citizens, streamlining service access, and alleviating the workload of municipal staff during critical situations. This study aims to know the following:

1. How effective will the mobile platform be in emergencies, such as natural disasters, in terms of informing and assisting citizens?

2. How can the mobile platform improve the delivery of essential government information and services to citizens of Basud?
3. What are the key functionalities that should be included in the platform to streamline communication and services between the local government and its citizens?

1.3 Significance of the Study

This study will provide research-based information on the effectiveness of implementing a mobile platform to enhance citizens' access to government information and services in the municipality of Basud. More specifically, this study will benefit the following individuals and organizations.

Basud Municipal Office. This study will assist all the departments in making their services more efficient and easily accessible to every citizen. The platform will serve as a centralized information hub for all important municipality announcements. Additionally, integrating government services into the platform will streamline the transaction process for each department, lessening the workload of all employees in each department by minimizing the need for in-person transactions in the office.

Municipal Risk Reduction Office (MDRRMO). The study will assist the MDRRMO in efficiently posting essential information before, during, and after a state of emergency such as earthquakes and tropical storms. Furthermore, the municipality's citizens can easily reach the office through the mobile platform without the fear of losing access to help and support when it is most needed.

Citizens of Basud Municipality. This study will help each citizen by providing an easier method of access to relevant information that the municipality provides and remote access to government services, lessening their burden instead of visiting the municipal office

in person. Wherein, each visit can take far more time than expected. The mobile platform will enable them to complete their task in the comfort of their own time using their mobile devices.

Business Owners. The study will help business owners stay updated with the latest and most accurate information that the municipality provides, reducing the likelihood of out-of-date information. This is particularly important for businesses, as they need to uphold permits and standards to continue running their businesses.

Other Researchers. This study will contribute to the existing body of knowledge that other researchers can use as reference material on the effectiveness of implementing technological initiatives in the transition from traditional government into digital government for further studies.

1.4 Objectives of the Study

This study aims to develop a mobile platform for efficient information and service dissemination in local government units. Furthermore, this study specifically aims to:

A. Develop a UI for the mobile application.
B. Develop a comprehensive mobile application that provides department-specific information.
C. Integrate key local government services into the mobile application and implement a notification system for updates.

1.5 Scope and Delimitations

This study focuses on creating a mobile phone application for enhancing both information and service dissemination for every citizen in the local government unit of Basud. This study will be narrowed down by the following:

A. The availability of integrated government services within the mobile application will depend on the permissions granted by each office and department of the local government of Basud.

B. External services that fall outside the jurisdiction of the Basud LGU, such as regional and national government services will not be included in this study.

C. The study will focus on evaluating the technical functionality of the mobile application and its immediate impact but will not delve into its long-term sustainability, budget considerations, and any policy reforms that the LGU may implement following the platform's deployment.

D. The testers for the mobile application will be limited to the locality of Barangay Poblacion II, Basud Camarines Norte.

E. This study will be limited to users who have access to smartphones and internet connections.

F. The study will not explore the impact of socio-economic factors on the adoption of the mobile platform, focusing instead on technological usability and service delivery.

G. The effectiveness of the mobile platform will be assessed based on user feedback gathered during the testing phase, which may limit the generalizability of the findings.

Chapter 2

REVIEW OF RELATED LITERATURE

2.1 Theoretical Background

2.1.1 Java

Java is a popular object-oriented programming language. It is a versatile language that can be applied to various developmental needs. The Java software platform is made up of the Java Virtual Machine (JVM), Java Application Programming Interface (API), and a complete development environment [8]. The Java bytecode is parsed, interpreted, and executed by the JVM. Web services, networking and security features, fundamental objects, and the creation of Extensible Markup Language (XML) are all included in the vast collection of libraries that make up the Java API [8].

Java is a popular object-oriented programming language that can run on various platforms such as computers, mobile devices, medical devices, and others. Java is still one of the most used languages for application software development [8]. An advantage of Java is that it is compatible across different platforms which means it can be run anywhere [9]. This study will utilize Java to develop a mobile application compatible with the Android OS.

2.1.2 Android

The Android operating system is one of the most popular mobile operating systems today, competing with the iOS created by Apple [10]. Android is developed by Google based on the Linux kernel and is designed for smartphones and tablets [11]. Android can run on various devices making it popular for different manufacturers. This operating system has been adapted for other devices including TVs, cars, smartwatches, smart home appliances,

etc. Applications installed in Android cannot access other app's data unless permitted by the user, making it a secure operating system [11]. This study will use Android OS as a platform for the application the researchers will develop.

2.1.3 Mobile Application

Mobile applications, commonly referred to as mobile apps, are software programs specifically designed to run on mobile devices such as smartphones and tablets. These apps cater to a wide range of functionalities, from productivity tools to entertainment, and have become integral to modern digital experiences. Mobile apps operate on platforms like iOS and Android, which provide the essential frameworks for app development and distribution [12].

The importance of mobile apps lies in their ability to simplify complex tasks and enhance accessibility to information and services. Their user-friendly interfaces and portability allow seamless interaction across diverse environments. Distribution through app stores, such as Google Play and Apple App Store, ensures users have secure and convenient access [12].

Research underscores mobile apps' economic and social impacts, demonstrating their role in innovation and sectoral growth, such as in business, healthcare, and education. They have enabled significant advancements in personalization, data management, and integration with AI and cloud computing, thus improving service efficiency and addressing modern challenges, such as digital transformation and effective information dissemination [13, 14].

2.1.4 Client Server Architecture

The client-server model is a fundamental framework in distributed computing, where distinct systems operate as clients and servers, connected over a network. Clients request

services or resources, and servers process these requests and provide responses. This architecture promotes efficiency, scalability, and resource centralization, often employed in databases, web services, and private information retrieval systems [15].

The client-server model also extends to cloud and serverless computing paradigms, enhancing scalability, cost-effectiveness, and fault tolerance by eliminating direct server management. [15].

2.1.5 Database

A database is an organized system for storing, retrieving, and managing data, essential in mobile application development for enabling functionality and enhancing user experience. It facilitates data storage for user information, app settings, and media, with lightweight options like SQLite being ideal for offline-first applications [16]. Moreover, security is paramount, with databases implementing encryption and access controls to protect sensitive information. Lastly, mobile-optimized databases integrate effectively with development frameworks, ensuring efficient workflows and compatibility. These features make databases a cornerstone of robust and scalable mobile applications [17].

2.2 Related Studies

2.2.1 Social Media and Information Dissemination

Social media platforms are online tools or services that let people connect while creating, sharing, and interacting with content. Through engagement with various media, individuals can cultivate networking, communication, amusement, and teamwork on these platforms. This study focuses on the use of social media platforms for the dissemination of information and creating awareness about library resources and services among students at Lagos State University. The results of this study showed that the major social media

platforms that are relevant to academic libraries are YouTube, WhatsApp, and Facebook. It also showed that social media provides a platform for lecturers to create chat rooms, forums, and groups to expand classroom discussion to post assignments, tests, and quizzes. The students indicated that the most used social media platforms are WhatsApp, Facebook, Twitter, Instagram, and YouTube. This shows that there are similarities as to which social media platforms are utilized and relevant to both academic libraries and students.

The main technology used in this study was the different social media platforms used by the students. This study focuses on the dissemination of information through various social media platforms. On the other hand, our study focuses on information dissemination through only one platform. Another difference is that this study's demographic is limited to students only while ours has a bigger target demographic [18].

2.2.2 Sustaining Digital Platforms in the Local Government

The utilization of technology in local government units (LGUs) is critical to improving data management and service delivery. In particular, the United Nations Development Programme (UNDP) developed DevLIVE+, a mobile and web-based platform that uses geospatial technology and data visualization tools to collect and manage household-level data. The platform was designed to assist LGUs in updating development plans, disaster preparedness, and responding to emergencies like the COVID-19 pandemic.

DevLIVE+ operates on an open-source framework, making it adaptable and scalable for different LGU needs. The technology supports real-time data collection, which enhances decision-making processes by providing accurate and timely information. Moreover, its integration of cloud computing and mobile data collection features allows for widespread deployment, particularly in remote areas where internet infrastructure may be limited.

Compared to DevLIVE+, the CitiServe platform adopts a broader focus, integrating essential government services into a mobile application to improve daily operations and citizen management. While DevLIVE+ emphasizes data collection and visualization for planning and emergencies, CitiServe prioritizes real-time communication, streamlined service access, and critical updates for citizens. CitiServe's localized implementation or the municipality of Basud ensures specialized solutions for its community, contrasting with the more generalized framework of DevLIVE+. CitiServe shifts the focus from primarily supporting LGUs to directly empowering citizens by centralizing information and services [19].

2.2.3 Digital Government: Mobile Applications and Their Impact on Access to Public Information

The exponential growth of mobile technologies worldwide has created opportunities for public organizations to enhance their digital governance capabilities. A recent study focused on developing a mobile application for Android to improve user access to municipal information in Cañete. The application specifically facilitated inquiries about procedures, driver's licenses, and municipal events. By leveraging mobile technology, the initiative sought to address the need for efficient, accessible public services.

The study utilized the Mobile-D methodology, a development framework emphasizing agility and quality. Its structured phases—exploration, initialization, production, stabilization, and testing—ensured timely development and usability of the application. The Android-based application demonstrated significant improvements in user access to information, as evidenced by pretest and posttest scores. The mean pretest score of 8.75 rose to 14.27 in the posttest, with a statistically significant difference of 9.05 points ($p < 0.05$).

These results validated the effectiveness of the mobile application in addressing its intended objectives.

While the referenced study focused on limited municipal functions, our research aims to develop a comprehensive mobile application for Local Government Units (LGUs). This application will provide centralized access to services and information from all municipal departments, including offices such as the Mayor's Office, Treasurer's Office, Civil Registry, Health Office, and Disaster Risk Reduction Office, among others. By integrating a wider array of municipal services, the proposed application seeks to enhance interdepartmental connectivity and improve overall user experience [20].

Chapter 3

METHODOLOGY

3.1 System's Model

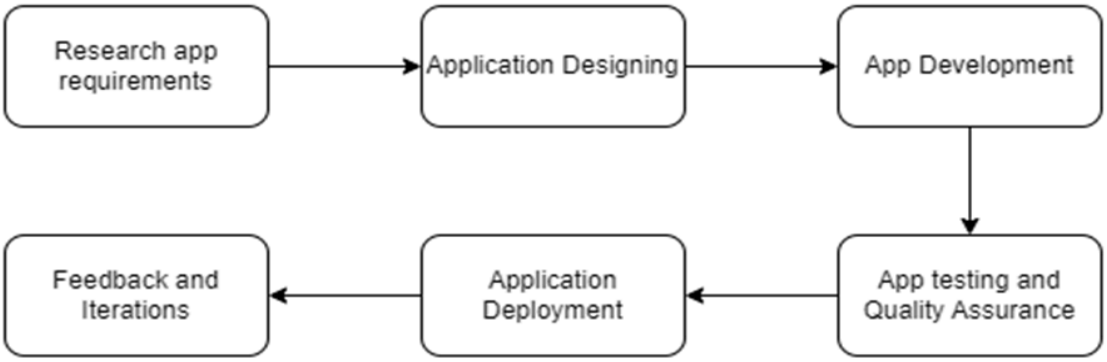

Fig. 1. Block Diagram for Creation of Application

A. Research App Requirements

The researchers will conduct a survey targeting the local residents in order to identify which government services they use most often. This survey will gather data on the different government services such as healthcare, education, permits, licenses, certifications, and more. This sector aims to understand the needs and preferences of the local community as well as specific government services that are highly demanded. The information gathered in this survey will serve as a basis for the prioritization of services to be included in the mobile application.

Following the survey, the researchers will prepare a formal request to the Local Government Unit (LGU) for permission to implement the identified government services in the mobile application. Additionally, the researchers will ask the LGU to provide the necessary information and guidelines required for users to avail of these services. This may

include documentation, application forms, eligibility criteria, contact information, and other relevant details.

B. Application Designing

The researchers will initiate the process by conceptualizing and designing the User Interface (UI) for the mobile application. This includes designing the app's general look and feel, as well as selecting color schemes, typography, and visual elements that are appropriate for the user's preferences and the application's function. The design process entails producing wireframes and mockups to visualize the layout and structure of each screen in the app. Optimizing efforts will also include ensuring that the application is responsive and works effectively on a variety of devices, including smartphones and tablets with varying screen sizes.

C. Application Development

The researchers will use Java as the programming language to develop the mobile application. Java is preferred for Android app development because of its platform independence, object-oriented characteristics, and broad library support, which make it an excellent choice for developing scalable and efficient applications. The researchers will focus on implementing the core functionalities of the application. This involves developing the app's architecture, writing the backend functionality, integrating any necessary APIs, and ensuring that the app's many components communicate seamlessly.

D. Application Testing and Quality Assurance

Initial Testing. After the mobile application is completed, the researchers will conduct preliminary testing to discover and fix any bugs or issues. This involves running the

program in a controlled environment and conducting multiple test cases to check that all functionalities work as expected.

Recruiting Testers. The researchers will assemble a group of early testers, who could be members of the planned target population, coworkers, or volunteers. These testers will provide helpful feedback on the application's functionality and usability.

E. Application Development

Once the program has been fully developed and tested, it will be released to major app stores including Google Play. This procedure involves creating the appropriate app listing information, such as the app description, screenshots, and advertising materials. The researchers will actively promote the software to potential users via various forms of media. This could involve social media campaigns, community engagement programs, and partnerships with local organizations and government authorities.

F. Feedback and Iterations

As users begin to use the app, the researchers will solicit comments to gain insight into user experience, satisfaction, and potential areas for development. This feedback is crucial for iteratively updating and improving the software.

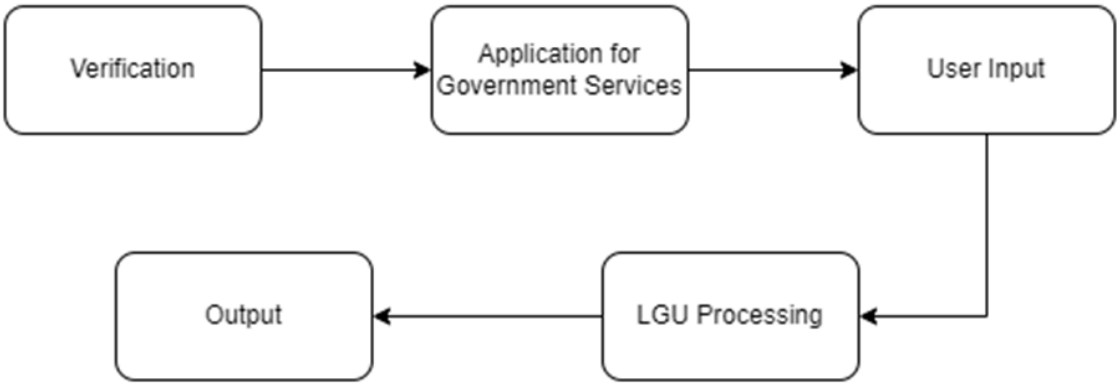

Fig. 2. Block Diagram of Application Process

A. Verification

When the user starts the app, they will be requested to input their login information. This usually includes a username and password. The login process guarantees that only authorized users have access to the app's features and services. Users must provide specified information during the initial registration or account setup process, such as their full name and permanent address. This information is required to validate the user's identity.

B. Application for Government Service

Once logged into the program, users will be given with a list of available government services. They can browse the list and choose the precise service they want to use, such as asking for a permit, seeking official documents, accessing healthcare services, or any other service offered by the local government. The app will provide full information and guidance on how to use each service, including required documentation, actions, and any associated fees. This helps consumers understand the procedure and ensures that they have all of the information needed to complete their requests.

C. User Input

When users select a specific government service to use, the program prompts them to submit the necessary information. This method guarantees that the required information is gathered to support the service request. As part of the identification process, users will be requested to confirm their full name and permanent address in order to verify their residency and eligibility for specific services. Users must upload a legitimate identification document, such as a government-issued ID, passport, or driver's license. This step helps to verify the user's identity and prevent fraudulent requests.

Depending on the service requested, users may be required to give extra information. For example, when applying for a business permit, users may be required to provide information about their company, such as its name, registration number, and type of operation.

D. LGU Processing

Request Processing. After submitting a service request using the application, the Local Government Unit (LGU) will receive it for processing. The LGU team will check the request's details to ensure that it is complete and fits the requirements for the service requested.

Verification Process. To guarantee the validity and correctness of the user's information, the LGU will compare the information provided in the request to their own records and databases. This verification process confirms the user's identity, eligibility, and any other criteria required for the service.

Validation and Approval. Once the verification procedure is completed, the LGU will validate and approve the request if all of the conditions are satisfied. The user will then be advised of the approval and any other steps they must take to fulfill the service request.

E. Output

Document Preparation. Once the LGU has processed and authorized the user's request, the appropriate document (such as a permit, certificate, or official record) will be created for release. The LGU team will ensure that the paper is completed correctly and meets all standards.

Notification. The user will be notified, either via the mobile application or another means of communication (such as email or SMS), when their requested document is ready for pickup. The message will provide information on where and how to collect the document.

Verification at Pickup. When the user arrives to get the document, they may be required to provide a valid ID or the notification they received to prove their identification. This step ensures that the paper is released to the appropriate person.

3.1.1 Program Flowchart

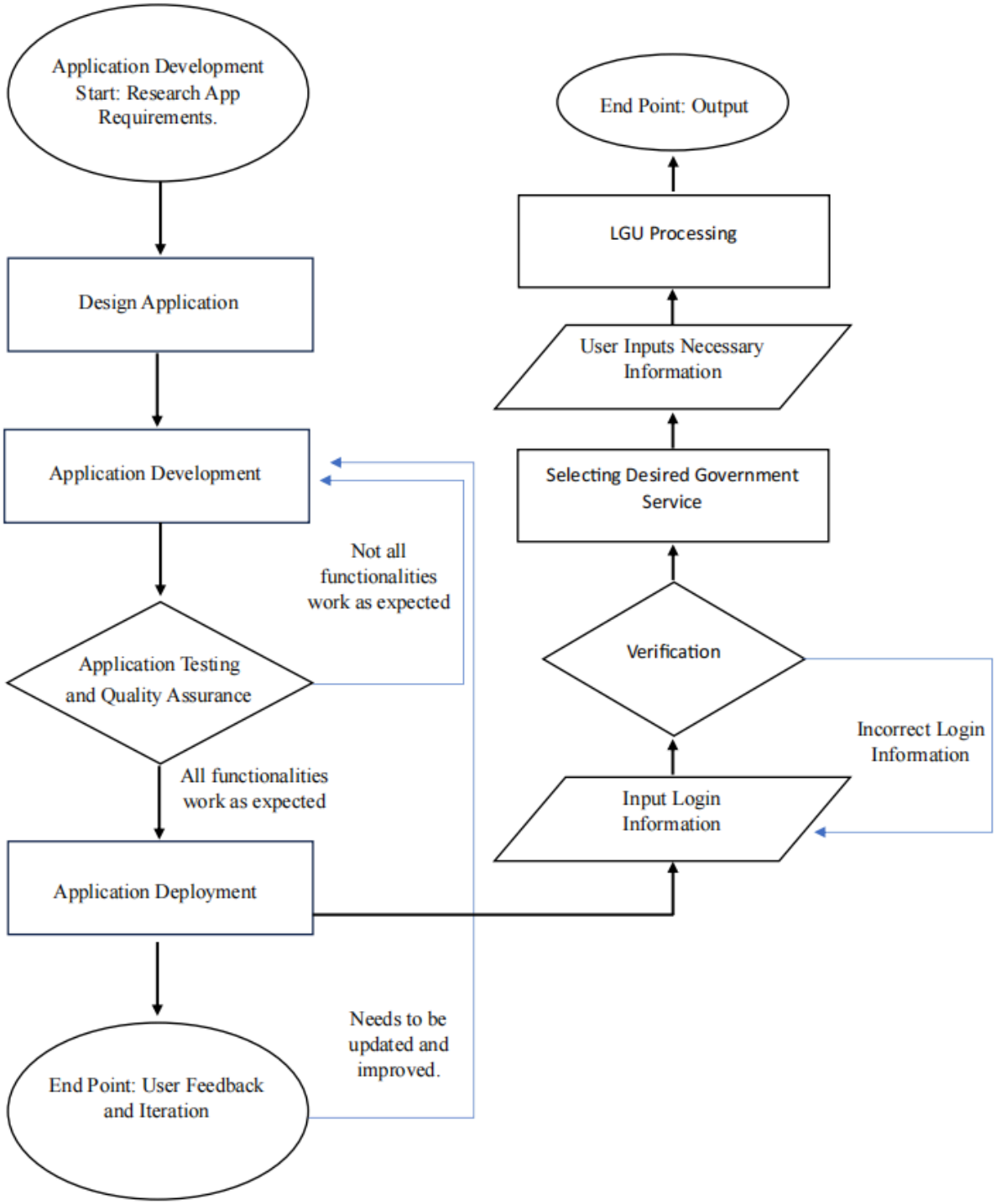

Fig. 3. Program Flowchart for both Application Development and Research Process

3.1.2 Mobile Application Mockup

Fig. 4. CitiServe Front Page

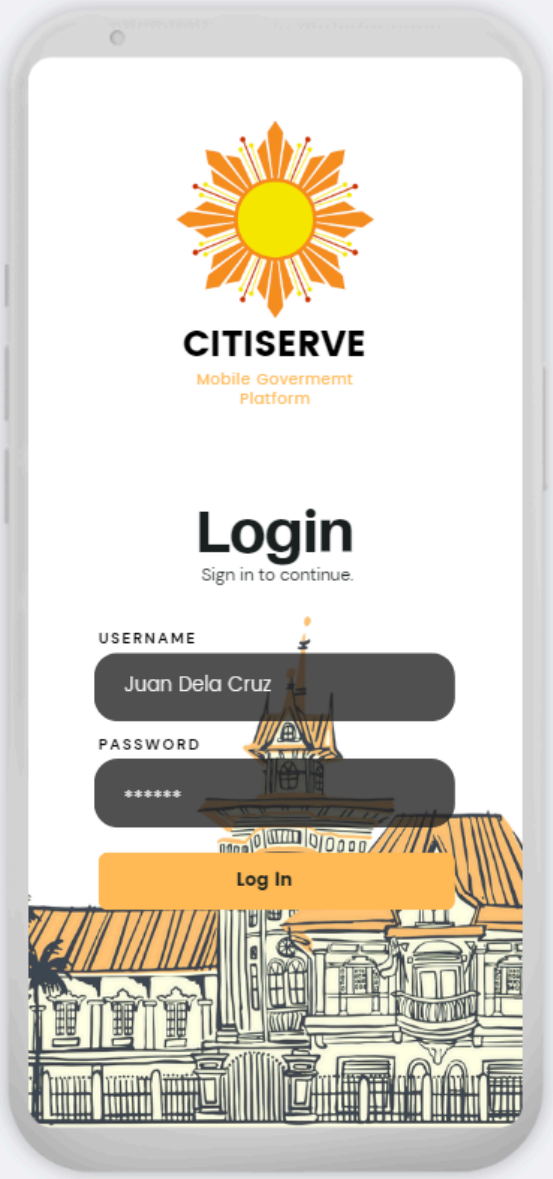

Fig. 5. CitiServe Login Page

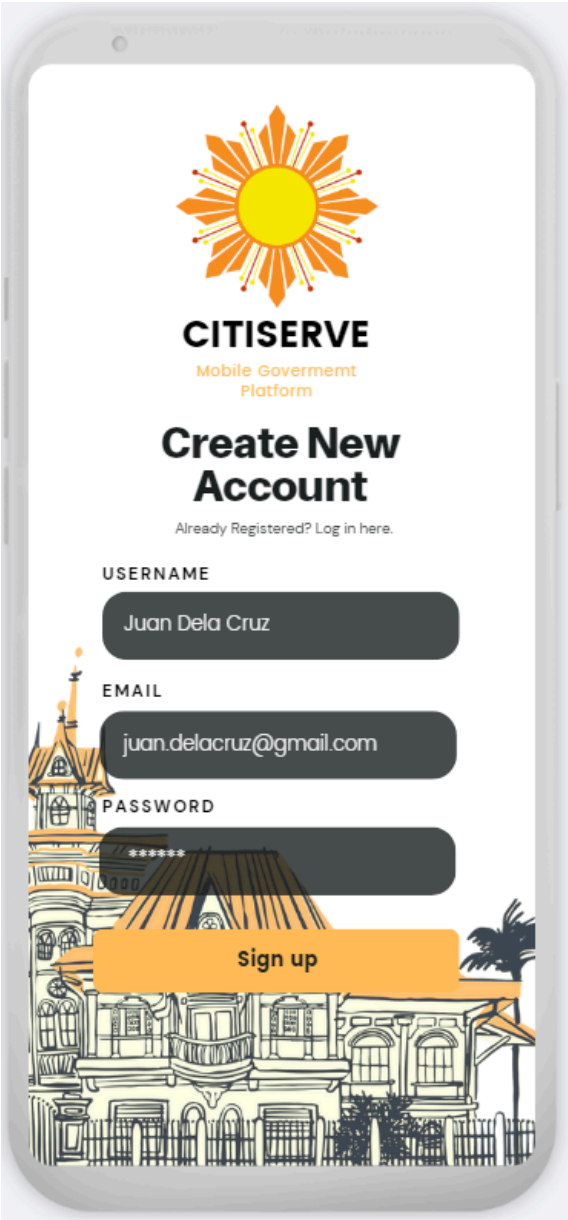

Fig. 6. CitiServe Sign Up Page

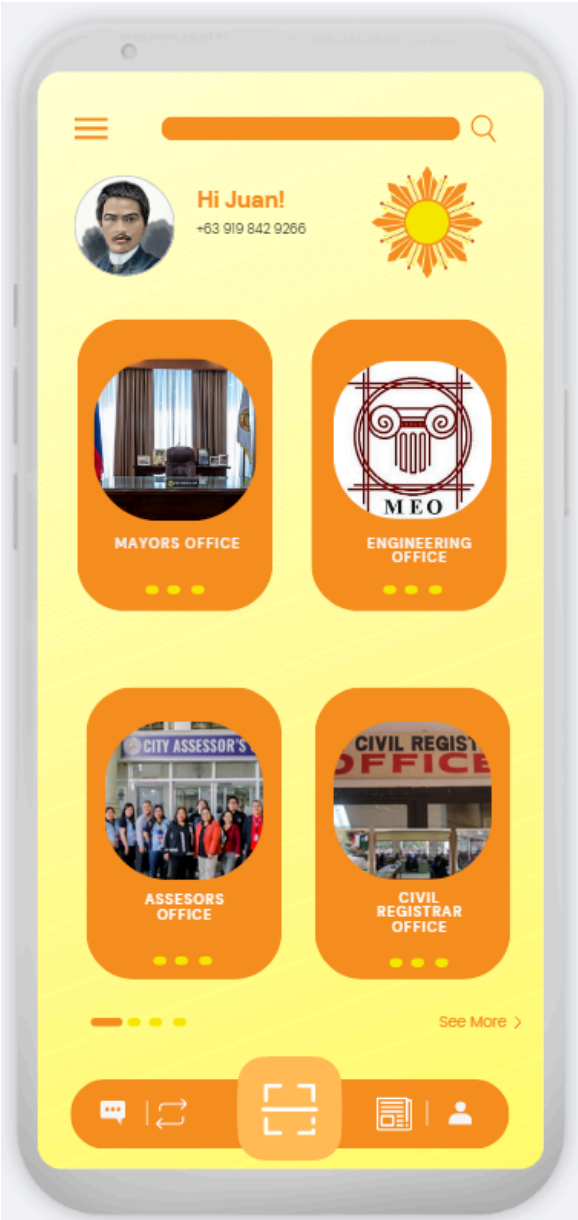

Fig. 7. CitiServe Home Page

3.2 Testing and Evaluation

The testing and evaluation phase of this research is critical in ensuring that the mobile platform is effective and meets the objectives outlined in the study. By collaborating with the local government unit of Basud, several respondents will be selected to be part of the beta

test. Furthermore, an iterative feedback mechanism will be adopted to ensure that the application aligns with the needs of the target users. Alongside a systematic approach to assess the platform's functionality, compatibility, performance, and security as outlined below:

Functionality Testing. This process will be conducted to ensure that all features on the mobile application are operating as intended. It involves the verification of the accessibility to each department-specific information, integration of local government services, and the reliability of the notification system. The respondents including local government staff and citizens of Basud will be tasked to interact with the mobile application to detect and resolve any bugs or inconsistencies. Tasks such as navigating the interface, finding specific pieces of information, and accessing all the integrated services provided by the LGU will be observed. Along with the collection of feedback through surveys and interviews from the respondents. Thus, detecting and resolving bugs or inconsistencies, ensuring both seamless and error-free operation of the platform.

Compatibility Testing. To ensure broad accessibility, the platform will be put to the test across various devices and operating systems. The application will undergo testing on smartphones and tablets with different screen sizes and resolutions, ensuring that it functions consistently across Android and iOS platforms. Any issues that arise related to device-specific hardware and software will be identified and resolved. Thus, guaranteeing a uniform user experience regardless of the user's device.

Performance Testing. The platform will be evaluated under varying conditions to measure speed, stability, and scalability. Response time for loading information and sending notifications will be analyzed. While stress test will be conducted to simulate peak usage

scenarios once it comes online to identify potential bottlenecks that may arise. These evaluations will ensure that the application remains both reliable and efficient, even during high-traffic periods.

Security Testing. As the application offers information and services concerning the local government unit, security is of utmost importance to safeguard user data and ensure secure transactions. Encryption methods for sensitive information will be assessed and authentication mechanisms such as passwords and biometric login will be tested for robustness. Moreover, vulnerabilities to injection attacks or data breaches will be identified through penetration testing. Thus, ensuring the platform adheres to high standards of security.

Chapter 5

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